

NAGO, Japan

WMO: 479400

Lat: 26.5942N

Lon: 127.9664E

Elev: 7

StdP: 101.24

Time Zone: 9.00 (E09)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.6	(c) 11.6	(d) 2.4	(e) 4.5	(f) 12.7	(g) 3.6	(h) 4.9	(i) 13.5	(j) 9.5	(k) 15.5	(l) 9.0	(m) 15.6	(n) 3.3	(o) 350	(p) 0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 4.4	(c) 32.5	(d) 26.7	(e) 31.9	(f) 26.5	(g) 31.4	(h) 26.4	(i) 27.7	(j) 30.6	(k) 27.4	(l) 30.3	(m) 27.1	(n) 30.0	(o) 4.0	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	22.7	29.4	26.6	22.2	29.3	26.2	21.7	29.2	88.5	30.7	87.0	30.3	85.6	30.2	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.4	(b) 8.8	(c) 7.8	(e) 8.6	(f) 33.6	(g) 1.2	(h) 0.7	(i) 7.7	(j) 34.0	(k) 7.0	(l) 34.4	(m) 6.3	(n) 34.8	(o) 5.4	(p) 35.3		
(a) 10.4	(b) 8.8	(c) 7.8	(e) 6.2	(f) 28.2	(g) 1.2	(h) 0.5	(i) 5.3	(j) 28.6	(k) 4.6	(l) 28.9	(m) 3.9	(n) 29.2	(o) 3.0	(p) 29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.0	16.4	17.1	18.4	20.9	24.0	27.0	29.2	29.2	28.1	25.3	22.0	18.2		
	DBStd	5.02	2.40	2.72	2.65	2.25	1.99	1.92	0.92	0.95	1.16	1.90	2.17	2.43		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	194	71	52	34	4	0	0	0	0	0	0	2	32		
	CDD10.0	4751	197	199	259	328	434	511	596	594	542	474	359	256		
	CDD18.3	1903	9	18	35	83	176	261	337	336	292	216	111	29		
	CDH23.3	17330	2	8	26	187	972	2562	4200	4163	3248	1581	351	30		
(14) Wind	WSAvg	3.8	3.8	3.8	3.8	3.5	3.2	3.9	3.9	3.7	3.7	4.2	3.8	3.9		
	PrecAvg	2035	98	113	137	165	225	263	173	235	232	167	120	106		
	PrecMax	3385	244	431	353	352	537	670	498	699	645	533	276	308		
	PrecMin	1334	19	25	31	34	21	63	9	47	10	37	11	28		
	PrecStd	465	52	83	71	91	123	143	135	156	161	113	70	62		
	DB	23.6	24.2	25.2	27.2	29.7	31.7	33.1	33.4	32.6	31.2	28.1	25.6	23.2		
	MCWB	20.1	20.9	21.7	23.7	25.8	27.1	26.7	26.9	26.4	25.6	23.7	22.3	20.4		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	22.4	23.2	24.0	26.1	28.6	31.1	32.4	32.6	31.7	30.0	27.0	24.2	21.6		
	2% MCWB	18.9	20.1	20.6	22.9	25.0	26.7	26.6	26.8	26.1	25.2	22.9	20.4	19.5		
	5% DB	21.4	22.4	23.2	25.2	27.8	30.4	31.9	32.0	31.1	29.1	26.1	23.1	20.3		
	5% MCWB	17.8	19.3	19.9	22.2	24.2	26.4	26.5	26.6	25.9	24.6	22.4	19.5	18.6		
	10% DB	20.4	21.5	22.4	24.4	27.0	29.8	31.3	31.4	30.5	28.1	25.4	22.2	19.2		
	10% MCWB	16.9	18.5	19.3	21.4	23.7	26.2	26.5	26.5	25.6	24.0	21.8	18.6	17.7		
	WB	21.1	21.7	22.5	24.7	26.7	27.9	28.1	28.2	27.7	26.7	24.7	23.2	21.6		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	22.9	23.7	24.3	26.3	28.5	30.6	30.8	31.1	30.8	29.2	26.7	24.8	22.3		
	2% WB	19.6	20.9	21.6	23.7	25.7	27.3	27.6	27.7	27.2	26.1	24.0	21.6	20.3		
	2% MCDB	21.6	22.8	23.5	25.5	27.7	29.8	30.5	30.7	30.1	28.8	26.2	23.3	21.6		
	5% WB	18.5	20.0	20.7	22.9	25.0	27.0	27.3	27.4	26.8	25.5	23.3	20.3	19.2		
	5% MCDB	20.7	21.8	22.6	24.8	27.1	29.4	30.3	30.4	29.7	28.2	25.6	22.5	21.7		
	10% WB	17.5	19.0	19.9	22.1	24.5	26.6	27.0	27.1	26.4	24.9	22.4	19.2	18.6		
	10% MCDB	19.9	21.2	21.9	24.0	26.5	29.0	30.1	30.2	29.3	27.6	24.9	21.7	20.4		
(35) Mean Daily Temperature Range	MDBR	4.9	4.9	5.2	5.1	4.8	4.0	4.4	4.7	4.9	4.7	4.7	4.7	4.8		
	5% DB MCDBR	6.0	5.6	5.5	5.2	5.1	4.2	5.0	5.4	5.4	5.3	5.3	5.3	5.8		
	5% DB MCWBR	4.1	3.9	3.9	3.2	2.7	1.8	1.8	1.9	2.1	2.3	3.0	3.9	4.8		
	5% WB MCDBR	5.2	4.8	4.9	4.5	4.4	4.0	4.4	4.6	4.7	4.6	4.4	4.8	5.8		
	5% WB MCWBR	4.3	4.1	4.4	3.4	2.7	1.9	1.8	2.0	2.1	2.4	3.0	4.1	5.8		
(40) Clear-Sky Solar Irradiance	taub	0.461	0.492	0.564	0.572	0.518	0.475	0.474	0.491	0.488	0.485	0.468	0.459	0.459		
	taud	2.116	2.052	1.887	1.926	2.142	2.329	2.330	2.269	2.246	2.194	2.194	2.129	2.129		
	Ebn at Noon	770	779	746	754	793	822	823	808	802	778	760	751	751		
	Edh at Noon	138	158	197	194	156	129	128	136	136	136	127	132	132		
(44) All-Sky Solar Radiation	RadAvg	2.39	2.99	3.74	4.39	4.86	5.34	6.53	5.96	5.21	4.06	2.94	2.40	2.40		
	RadStd	0.29	0.45	0.36	0.39	0.64	0.48	0.52	0.39	0.37	0.34	0.28	0.23	0.23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.75	+1.01	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	+0.41	+0.53	N/A	N/A	N/A	N/A

Nomenclature: See separate page